



Fusion at multiple radii: a rotation-invariant uniform LBP for finger-vein identification

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Received: 6 September 2025 / Revised: 18 November 2025 / Accepted: 21 November 2025 / Published online: 5 December 2025
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Abstract

This study proposes a novel Multi-Radius Fused Rotational Invariant Uniform Local Binary Pattern variant for finger vein identification, specifically designed for systems with limited computational resources. The proposed operator, $LBP_{(8,1),(16,1),(8,2)}^{riu2}$, integrates three distinct radius configurations within a unified sliding window framework to capture multi-scale textural information and enhance discriminative capability. This texture-based feature extraction technique is systematically evaluated in combination with dimensionality reduction methods, including Principal Component Analysis (PCA), Two-Dimensional PCA (2DPCA), and Two-Directional Two-Dimensional PCA (2D²PCA), to assess its robustness across diverse operational scenarios. Comprehensive experiments are conducted on the FV-USM, MMCBNU-6000, and UTFVP datasets. Performance evaluation is carried out using two fusion strategies across three standard experimental protocols. Results demonstrate that the proposed method, along with its dimensionality-reduced variants, achieves competitive performance and outperforms traditional hand-crafted techniques. Furthermore, comparative analyses with the state-of-the-art approaches confirm the effectiveness of the proposed texture-based method. This research establishes the practical performance boundaries of classical feature extraction techniques, with results rigorously validated using Cumulative Match Characteristic (CMC) curves.

Keywords Finger vein identification · Biometrics · Feature extraction · Fusion · Dimensionality reduction · Local Binary Patterns (LBP)

1 Introduction

The demand for secure identification in biometrics has exposed the weaknesses of traditional methods. Among the various modalities, vascular biometrics stand out for their stability, resistance to spoofing, and inherent liveness detection [1]. Finger vein recognition (FVR) leverages near-infrared (NIR) imaging to capture unique and stable subcutaneous vascular patterns, making it well-suited for high-security applications. Early FVR research relied on vein-line segmentation, but its error sensitivity led to the adoption of

texture-based descriptors, notably the Local Binary Patterns (LBP). Since its first application on FVR in 2009 [2], LBP has been extended for robustness against noise, rotation, and contrast. However, single-radius LBPs still lack discriminative power under low contrast or high inter-class similarity, motivating multi-scale and hybrid approaches. Meanwhile, Convolutional Neural Networks (CNNs) have emerged as strong alternatives for high-complexity systems, yet LBP remains attractive for its efficiency, interpretability, and suitability in resource-constrained environments.

This study proposes a novel Multi-Radius Fused $LBP_{(8,1),(16,1),(8,2)}^{riu2}$ operator for finger-vein identification, where the choice of the radii is informed from prior work. The descriptor's robustness is evaluated through integration with Principal Component Analysis (PCA), Two-Dimensional PCA (2DPCA), and Two-Directional Two-Dimensional PCA (2D²PCA) for dimensionality reduction, with comprehensive testing on the FV-USM, MMCBNU-6000, and UTFVP datasets. A high-level system workflow is provided in Figure 1. The main contributions of this paper are:

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